(Please write your Enrollment Number)

Enrollment No.

End-Term Examination
(B. Tech CSE/ECE CBCS) (SUBJECTIVE TYPE)

(April, 2022)

Subject Code: BAS 105	**Subject: Applied Chemistry**	
Time : 3 Hours		**Maximum Marks : 60**
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.		
Q1.		
	(a) Differentiate between autocatalysis and auto catalytic poisoning giving suitable examples?	**(2.5*8=20)**
	(b) The absorbance of an iron thiocyanate solution containing 0.00300 mg Fe/mL was reported as 0.2900 at 540 nm. Calculate the specific absorptivity, including units, of iron thiocyanate on the assumption that a 1.00 cm cuvette was used.	
	(c) What are the advantages of discontinuous aligned fibers over continuous aligned fibers?	
	(d) What is sol gel method of preparation of metal oxide nanoparticles?	
	(e) In the phase diagram of ice, the fusion curve of ice has negative slope. Explain.	
	(f) Why Calgon conditioning is better than phosphate conditioning? Explain with the help of chemical reactions.	
	(g) What is the effect of heating rate and furnace atmosphere on TG curve? Discuss the potential applications of TGA in qualitative analysis for the decomposition of calcium oxalate.	
	(h) A completely exhausted zeolite softener requires 120 liters of NaCl solution having 100g/L of NaCl. How many liters of water having hardness 500 ppm can be softened by the zeolite.	
	UNIT-I	
Q2.	(a) A 50 mL of hard water containing 1.2 g of $CaCO_3$ per liter required 15 mL of EDTA solution for the end point. 50 mL of water sample required 19 mL of EDTA solution and 50mL of boiled and filtered sample required 11 mL of EDTA solution for the end point. Calculate temporary and permanent hardness of water sample in ppm. (b) Explain why alkalinity of water cannot be due to simultaneous presence of OH^-, CO_3^{2-} and HCO_3^-? (c) What is blow down process?	**(6,2,2)**
Q3.	(a) A water sample contains Ca $(NO_3)_2$ = 164 mg/L, Mg $(HCO_3)_2$ = 73 mg/L, $CaCl_2$ =222 mg/L, $MgSO_4$=120 mg/L. Calculate the amount of Lime (74 % pure) and Soda (90 % pure) required for softening 5000 liters of hard water (b) Why magnesium bicarbonate requires double amount of lime for softening?	**(8,2)**
	UNIT-II	
Q4.	(a) What is the advantage of catalyst over high temperature to promote chemical reactions? (b) How silver is profitably recovered from the Argentiferous lead? Draw and explain this process with a suitable phase diagram?	**(3,7)**
Q5.	(a) What are acid base catalysts? Discuss the mechanism of acid catalyzed reaction with suitable example. (b) Explain what do you understand by the following terms (i) Congruent melting point (ii) Eutectic point	**(6,4)**

	UNIT-III	
Q6.	(a) What is Thermal analysis? (b) Explain the basic principle and instrumentation of UV-Visible spectrophotometer (c) What are the limitations of Beer-Lambert's law?	
Q7.	(a) Why only n $\longrightarrow \pi^*$ and $\pi \longrightarrow \pi^*$ transitions are useful in interpretation of UV spectra? (b) Explain the basic principle and instrumentation involved in DTA. Why TGA is always associated with DTA?	
	UNIT-IV	
Q8.	(a) Discuss the top down and bottom up approaches of nanomaterials synthesis. Which one is better according to you and why? (b) Write a note on Carbon Nanotubes along with its application.	
Q9.	What are the factors that need to be considered in making Fiber-reinforced composites and structural composites? Explain with suitable diagram	